HEAPLINE · CONCEPT OPTIONS

HeapLine concept options

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DOCUMENT CONTROL

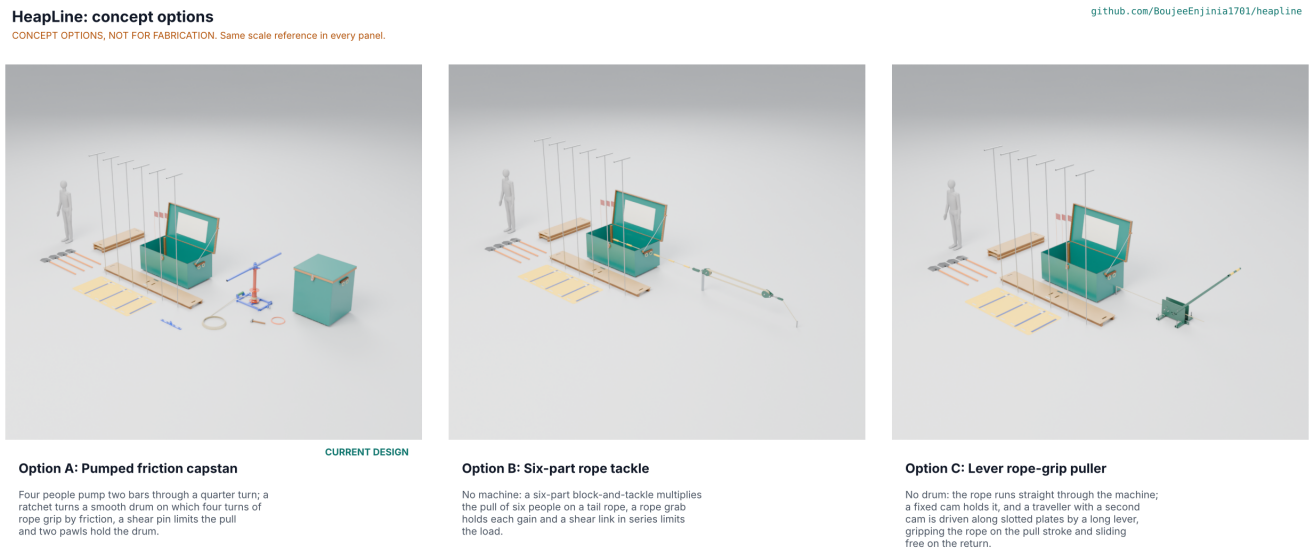
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First options study, written for the kit 1.9.1 upgrade; option A is the current design, options B and C added; recommendation Proposed, awaiting Amish

HeapLine concept options

Option A (the pumped friction capstan, the current design) scores 90 of 125 (72.0 %) and option B (a six-part rope tackle) 87 (69.6 %): the two are within 3.3 %, inside the 10 % band, so both go to Amish without a pick, and A stays the design. Option C (a lever rope-grip puller) scores 64 (51.2 %). The scores are the author's first-pass judgement against the user requirements, not measurements. Proposed, awaiting Amish. Method: STANDARDS section 21.



The three options differ only in the device that pulls sheeting, timber or debris off a mark; the search kit (crate, boards, probes, shovels, stretcher, lookout kit) is common to all and is the same in every model. All three are modelled to detail level D3 in `cad/src/model.py` and pass `python .kit/cad_check.py --export`. No option that Amish already declined is proposed again: HPL-DDR-001 item 1 declined the SiltHaul winding drum with chain drive and the walk-round capstan, items 6 and 7 fixed what the capstan may pull and that probes are never driven, and decisions 21B, 22B, 22C and 23A and 23B (a shoulder-loop haul line, lighter boards, a handcart, sharing alone, a reused crate alone) were not chosen. A crank-driven winding drum was therefore dropped from the options at this session; see HPL-DDR-005.

The options

Option A: pumped friction capstan (current design)

Working principle: four people pump two bars through a quarter turn; a drive ratchet turns a smooth drum on which four turns of 14 mm rope grip by friction (capstan equation), a 6 mm shear pin limits the pull to about 6.4 kN and two pawls hold the drum. It gives 5 kN with 114 N each from four people at 31 W each (S2, S3); the pull set is 113 parts in a 347 part model, the shared capstan set costs USD 547 with its crate, and the heaviest item is 20.9 kg. Key risks: the shear pin spread leaves 2 % margin to 5 kN (S6) and the capstan needs a sound anchor. It needs from a co-design partner: a sound tree, vehicle or column near the slide, and the drill.

Option B: six-part rope tackle

Working principle: no machine. Two triple blocks reeved six-part multiply the pull of the people on a tail rope; a rope grab holds each gain; a shear link in series with the load limits the pull. With a sheave efficiency of 0.93 (estimate) the real advantage is 4.69, so 5 kN needs 1,066 N on the tail: six people at 178 N each (S21, after the six-handle fix

of HPL-DDR-006). The pull set is 22.5 kg and about USD 306 (estimate). Key risks: only about 1.1 m of travel before the blocks meet and the grab resets; two anchors (a picket and a load stake); the grab hold is an estimate (S22). It needs: six people at the tail, which is more than the capstan wave.

Option C: lever rope-grip puller

Working principle: no drum. The rope runs straight through the machine; a fixed cam holds it and a traveller carrying a second cam is driven along slotted plates by a 1.6 m lever, gripping on the pull stroke and sliding free on the return (Tirfor-like in principle, here for fibre rope). Ratio 12.2, efficiency 0.7 (estimate): 146 N each from four people at 54 W each (S24, S25, after the 1.6 m lever fix). The pull set is 78 kg and about USD 453 (estimate). Key risks: the cams must clamp the rope at about 8.5 MPa, which may glaze or cut polyester (S26), and there is no overload device yet. It needs: rope supplier data and a grip test.

First-order numbers

Table 1. Options compared.

Quantity	Option A	Option B	Option C
Estimated parts cost, shared capstan set with crate (USD)	547 (BOM)	306 (estimate)	453 (estimate)
Mass of the pull set (kg)	51.7	22.5	78.0
Output	5 kN, 1.5 m/min	5 kN, about 1.1 m a reset	5 kN, 1.8 m/min
Effort	114 N each, four people, 31 W each	178 N each, six people	146 N each, four people, 54 W each
Parts in the D3 model of the whole kit (pull set)	347 (113)	236 (66)	217 (47)
Mass of the D3 model of the whole kit (kg)	293.9	186.0	241.5
STEM check	9 within, 14 plausible (S1 to S20 and S27 to S29, shared with the kit)	0 within, 3 plausible (S21 to S23)	0 within, 3 plausible (S24 to S26)

Costs of B and C are estimates scaled from BOM line 9 (USD 62 for 19.2 kg of welded steel) plus the rope, slings and crate lines; they are not quotes. Numbers come from `docs/04-calcs/options_calc.py` and HPL-CAL-001.

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5).

Criterion	Source	Weight	A	B	C	Reason
Pulls 5 kN with the people who arrive in the capstan wave (four at the bars)	UR-004	3	5	2	3	A: 114 N each, four people. B: 178 N each needs six people (S21). C: 146 N each, four people, but the lever is 1.6 m long (S24)
Overload limit and holds when let go	UR-010	3	4	3	2	A: shear pin and two holding pawls (R10, R11). B: a shear link in series and a rope grab, but the grab hold is an estimate (S22). C: no overload device yet, and a released lever swings
Carried by four people or fewer, nothing too heavy	UR-008	3	3	5	2	Pull set from the D3 models: A 51.7 kg (heaviest item 20.9 kg), B 22.5 kg, C 78.0 kg with rails and a 1.6 m lever
Cost against the value-engineering target	UR-009	3	3	5	4	A USD 547 with crate (BOM). B about USD 306 and C about USD 453, both estimates scaled from BOM lines (options_calc.py)
Set up in the first minutes	UR-001	3	3	4	3	A: stakes, sling and rope. B: picket, two blocks reeved, load stake. C: four stakes, rails, rope threaded through the cams
Speed and reach	UR-004	2	4	2	4	A 1.5 m/min continuous. B at most about 1.1 m of load travel before the blocks meet and the grab resets. C 1.8 m/min
Buildable with local skills and tools	UR-011	2	3	3	2	A: welded frame, profile-cut ratchets. B: plate and cast or turned sheaves. C: machined cams and slotted plates
Maintenance and repair in the field	UR-011	2	3	4	2	B has the fewest parts and no ratchets; C's cams wear the rope and need re-cutting
Rope life and storage	UR-007	2	4	3	1	A: four gentle turns on a 114 mm drum (S9). B: bend ratio 8.9 (S23). C: 8.5 MPa clamp on the rope (S26)
Confidence from the STEM check	HPL-STM-001	2	4	3	2	A: 9 within and 14 plausible rows, each closed by a test, a datasheet or a purchase specification (the pin bar, S5 to S7). B: S21 to S23 plausible, all on estimates. C: S24 to S26 plausible, with no overload device
Weighted total (maximum 125)			**90**	**87**	**64**	A and B within 3.3 %

Recommendation

No pick between A and B: they are within 10 %, so both go to Amish (HPL-DDR-005, Proposed, awaiting Amish). A is the current design and stays the design unless Amish decides otherwise. B wins on mass, cost and field repair and loses on the people it needs and its short reach per reset; its best idea to graft onto A is a rope grab at the tail, as a second way to hold the load when the drum is let go. Option C scores lowest on this scoring, mainly on mass, rope life and the lack of an overload device; it is not recommended.